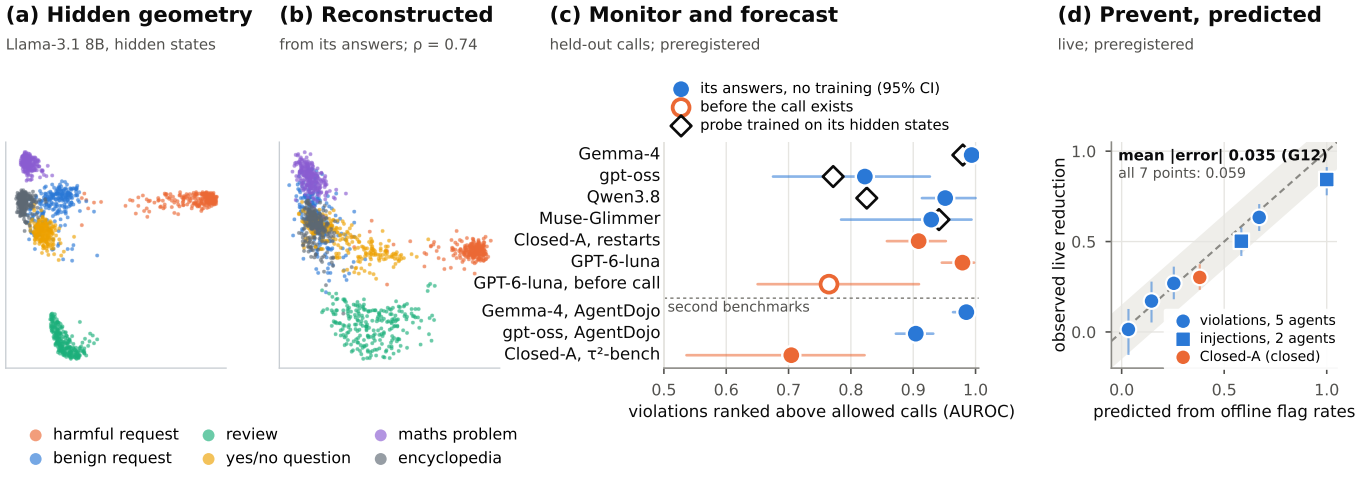

(a) Hidden geometry
Llama-3.1 8B, hidden states
(b) Reconstructed
from its answers; ρ = 0.74
(c) Monitor and forecast
held-out calls; preregistered
its answers, no training (95% CI)
before the call exists
probe trained on its hidden states
Gemma-4
gpt-oss
Qwen3.8
Muse-Glimmer
Closed-A, restarts
GPT-6-luna
GPT-6-luna, before call
second benchmarks
Gemma-4, AgentDojo
gpt-oss, AgentDojo
Closed-A, τ²-bench
0.5
0.6
0.7
0.8
0.9
1.0
violations ranked above allowed calls (AUROC)
(d) Prevent, predicted
live; preregistered
mean |error| 0.035 (G12)
all 7 points: 0.059
violations, 5 agents
injections, 2 agents
Closed-A (closed)
observed live reduction
predicted from offline flag rates
0.0
0.5
1.0
harmful request
benign request
review
yes/no question
maths problem
encyclopedia